

Product Name: **2-HYDROXYETHYL METHACRYLATE**

1. Chemical product and company identification

Product name: 2-HYDROXYETHYL METHACRYLATE

Name of the supplier: NIPPON SHOKUBAI CO., LTD.
 Address: Kogin Bldg., 4-1-1 Koraibashi, Chuo-ku, Osaka 541-0043 Japan
 Charge section: Responsible Care Division
 (TEL:81-6-6223-9166, FAX:81-6-6202-1766, E-MAIL:MSDS_NSCL@shokubai.co.jp)

 Emergency phone number: Himeji Plant Environmental & Safety Department
 (TEL:81-79-273-1139, FAX:81-79-274-3723)
 Recommended use: Paints, Adhesives, Printing inks
 Restrictions on use: No information

2. Hazard identification

GHS classification of product	[Category]
Flammable liquids	Not classified
Acute toxicity (oral)	Not classified
Acute toxicity (dermal)	Not classified
Skin corrosion/irritation	Not classified
Serious eye damage/eye irritation	Category 2
Skin sensitization	Category 1
Acute hazards to the aquatic environment	Not classified
Chronic hazards to the aquatic environment	Not classified

GHS label elements

Pictograms:



Signal word: Warning
 Hazard statement: Causes serious eye irritation
 May cause an allergic skin reaction

Precautionary statements:

Prevention Avoid breathing dust/fume/gas/mist/vapours/spray.
 Wash hands and face thoroughly after handling.
 Contaminated work clothing should not be allowed out of the workplace.
 Wear protective gloves/protective clothing/eye protection/face protection.

 Response Take off contaminated clothing and wash it before reuse.
 IF ON SKIN: Wash with plenty of water.

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If skin irritation or a rash occurs: Get medical advice/attention.

If eye irritation persists: Get medical advice/attention.

IF IN EYES: Rinse cautiously with water for several minutes.

Remove contact lenses, if present and easy to do. Continue rinsing.

Disposal Dispose of contents/container in accordance with local/regional/national/international regulation.

Other hazards: The product can be charged with static electricity.
Even though a polymerization inhibitor is added, polymerization can be initiated by heating, direct sunlight, peroxides, and iron rust. Progress of polymerization can cause temperature increase and accelerated vapor pressure rise, resulting in explosion.

3. Composition/information on ingredients

Substance or mixture: Substance

Chemical name	Content wt. %	CAS number
2-Hydroxyethyl methacrylate (Inhibited)	97-100	868-77-9
2-(2-Hydroxyethoxy) ethyl methacrylate (Inhibited) (impurity)	0-2.8	2351-43-1

4. First-aid measures

Inhalation: Remove person to fresh air and keep comfortable for breathing.

Immediately call a POISON CENTER/doctor.

Skin contact: Immediately wipe away the attached material with cloths and the like rapidly.

Wash with plenty of soap and water.

If skin irritation or a rash occurs: Get medical advice/attention.

Eye contact: Wash the eyes with clean water for at least several minutes (desirably 15 minutes or longer), (Remove the contact lenses if possible.) and get medical attention by an eye specialist immediately. During washing, fully open the eyelid with fingers and wash in such a way as to let water reach to every corner of the eyeball and eyelid.

Ingestion: Call a POISON CENTER/doctor if you feel unwell.

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5. Fire-fighting measures

- Suitable extinguishing media: Small Fire: Chemical powder extinguisher, alcohol resistant foam fire extinguishing media, carbon dioxide or water spray.
Large Fire: Alcohol resistant foam fire extinguishing media, water spray or fog.
- Unsuitable extinguishing media: Direct straight streams (may spread a fire)
- Specific hazards of fire-fighting: Will be easily ignited by heat, sparks or flames.
Containers may explode when heated. Fire will produce irritating, corrosive and/or toxic gases. May polymerize explosively when heated or involved in a fire. Vapor explosion hazard indoors, outdoors or in sewers.
- Specific extinguishing methods: Keep upwind of fire.
Use of water spray at a large fire, when another fire extinguishing media may be inefficient.
Fight fire from maximum distance or use unmanned hose holders or monitor nozzles. Cool containers with flooding quantities of water until well after fire is out. For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn.
- Special protective actions
for fire-fighters: In case of fire fighting, wear positive pressure self-contained breathing apparatus (SCBA) and chemical protective clothing. Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.

Structural firefighters' protective clothing will only provide limited protection.
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6. Accidental release measures

- Personal precautions, protective equipment and emergency procedures: Promptly remove possible ignition sources from the vicinity.

Use personal protection recommended in Section 8. Isolate the hazard area and deny entry to unnecessary and unprotected personnel.
- Environmental precautions: Do not flush to sewer or allow to enter waterways.
This product has a strong bad smell and irritation. Therefore, take proper measures like informing the people in the neighboring area of the leakage occurrence.
Surround this product with sandbags and earth/sand and cover it with an antistatic sheet to prevent diffusion (of smell).

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Methods and materials

for containment and cleaning up: In the case of small amount, hydrolyze it with 5 - 10% sodium hydroxide solution, neutralize, and wash with water. The waste water should be properly treated (incineration). In the case of large amount, surround it with earth/sand, sandbags and others, cover it with oil-resistant antistatic sheet to prevent generation of vapor, and recover it into proper containers like drums. Then, treat it as in the case of small amount.

7. Handling and storage
Handling

Technical measures: This product can be charged with static electricity. Take countermeasures for static electricity removal (grounding, others). Wear antistatic clothes and antistatic shoes to prevent human body electrification.

Use explosion-proof [electrical/ventilating/lighting] equipment.

Use non-sparking tools.

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

Do not handle the containers harshly in such a way as to make them tumbled, dropped, given a shock, or dragged.

Local/total ventilation: Use the ventilation equipment described in Section 8.

Precautions for
safe handling: Not especially.

Contact avoidance: Refer to the information in section 10 (Stability and reactivity).

Hygiene measures: During handling, be sure to wear proper protective equipment (refer to the section 8).

Storage

Conditions for
safe storage: Keep away from heat.
Protect from sunlight. Store in a well-ventilated place.
Keep away from sources of ignition. No smoking.
When storing this product in a tank, store it at as-low-as-possible temperature of 40°C or lower and control to keep the oxygen concentration of the gaseous phase at 7 - 21%. (Storing at oxygen concentration of lower than 7% can cause polymerization.)

Safe packaging material: Store in tightly closed container.

Stainless steel or polyethylene container

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8. Exposure controls/personal protection

Permissible concentration

Chemical name	ACGIH		U. S. -OSHA	
	TLV-TWA	TLV-STEL	PEL-TWA	PEL-STEL
2-Hydroxyethyl methacrylate (Inhibited)	Not established	Not established	Not established	Not established
2-(2-Hydroxyethoxy) ethyl methacrylate (Inhibited) (impurity)	Not established	Not established	Not established	Not established

Appropriate engineering controls: Install eye and body washing facilities near the handling place.

Use closed equipment/machinery or use local ventilation equipment.

Personal protective equipment

Respiratory protection: Use appropriate respiratory protection.

Gas mask

Consider wearing an air-supplied respirator when handling highly concentrated chemicals.

When using a gas mask in an environment where workers are exposed to dust, use a combined filter for both hazardous gas and particle.

Select a gas mask with performance and structure suitable for the work that complies with the laws and standards of the area where it is handled, referring to the data provided in the instruction manual.

Hand protection:

Use appropriate protective gloves.

Organic solvent impermeable protective gloves (Antistatic ones are desirable.)

For the selection of protective gloves, set a sufficient usage time for work by referring to the class of permeability described in the instruction manual, and use gloves within that time.

Eye/face protection:

Use appropriate protective eyewear.

Protective glasses, goggle, protective face shield

Skin and body protection:

Use appropriate protective clothing and shoes.

Organic solvent impermeable protective clothes, protective shoes (Antistatic ones are desirable.)

9. Physical and chemical properties

[Chemical product]

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Physical state:	Liquid
Form:	Liquid
Colour:	Clear and colorless
Odour:	Ester odor
Melting point /freezing point:	<-10(°C)
Boiling point or initial boiling point and boiling range:	87(°C) (667Pa)
Flammability:	No data
Lower and upper explosion limit /flammability limit:	No data
Flashpoint:	110(°C)Cleveland Open Cup
Auto-ignition temperature:	367(°C)
Decomposition temperature:	No data
pH:	4.3(20°C、1mol/L Aqueous solution)
Kinematic Viscosity :	No data
Solubility:	water : Completely soluble
Partition coefficient n-octanol/water (log value):	0.42
Vapour pressure:	16.8(Pa) (25°C)
Density and/or relative density:	1.070(25°C)
Relative vapour density:	No data
Particle characteristics:	No data
Other data:	Conductivity:6.6E-06 S/m(25°C) Heat accumulation storage test(UN recommendations) SAPT > 75°C (SAPT : Self-accelerating polymerization temperature)

10. Stability and reactivity

Reactivity:	Refer to the information in "Possibility of hazardous reactions".
Chemical stability:	Even though a polymerization inhibitor is added, heating, direct sunlight, peroxides, and iron rust can cause polymerization.
Possibility of hazardous reactions:	When storing at low-oxygen condition of oxygen concentration of lower than 7% at the gaseous phase, polymerization can occur.
Conditions to avoid:	Heating, contact with incompatible substances, ignition sources
Incompatible materials:	Strong oxidizing substances, peroxides.

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Hazardous

decomposition products: No information

11. Toxicological information

[Chemical product]

Acute toxicity, oral:	Not classified LD50:> 4000 mg/kg[rat] LD50: 5050 mg/kg[rat]
Acute toxicity, dermal:	Not classified LD50:> 3000 mg/kg[rabbit]
Acute toxicity, inhalation:	Not classified (Gas) Classification not possible (Vapour) Classification not possible (Dust/Mist) No test data on the product
Skin corrosion/irritation:	Not classified Slightly [rabbit] (Tested by OECD Test Guideline[404] as this product.)
Serious eye damage/irritation:	Category 2
Sensitization, respiratory:	Classification not possible No test data on the product
Sensitization, skin:	Category 1 Japan Society for Occupational Health Skin Sensitization Group 2
Germ cell mutagenicity:	Classification not possible No data available
Carcinogenicity:	Classification not possible No test data on the product
Reproductive toxicity:	Classification not possible No data available
Specific target organ toxicity (Single exposure):	Classification not possible No data available
Specific target organ toxicity (Repeated exposure):	Classification not possible No data available
Aspiration hazard:	Classification not possible No test data on the product

12. Ecological information

[Chemical product]

Ecotoxicity

Acute hazard: Not classified

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Fish:	LC50(96hr): > 100 mg/L[Oryzias latipes] LC50(96h): 227 mg/L[Pimephales promelas] LC50(96h): 213-242 mg/L[Pimephales promelas]
Crustacea:	EC50(48hr): 380 mg/L[Daphnia magna]
Algae or other aquatic plants:	ErC50(72hr): 710 mg/L[Pseudokirchneriella subcapitata]
Chronic hazard:	Not classified
Fish:	No test data on the product
Crustacea:	NOEC(21day): 24 mg/L[Daphnia magna]
Algae or other aquatic plants:	No test data on the product
Persistence and degradability:	Rapidly biodegradable
Bioaccumulative potential:	No bioaccumulation
Mobility in soil:	No test data on the product
Hazardous to the ozone layer:	Classification not possible No test data on the product

13. Disposal considerations

Disposal methods: When waste material and waste water are to be treated, incinerate them or collect them into specified containers and entrust the disposal to a licensed industrial waste disposal company.

Do not use the used containers for other purposes like filling other substances. After treating the content according to the above description, the empty containers should be reused, recycled, or disposed through suitably qualified or licensed contractor, in accordance with national/local regulations.

14. Transport information

ICAO/IATA	UN classification :	Not regulated as a dangerous good
	UN number :	Not regulated as a dangerous good
	Proper shipping name :	Not regulated as a dangerous good
	Packing group :	Not regulated as a dangerous good
IMDG	UN classification :	Not regulated as a dangerous good
	UN number :	Not regulated as a dangerous good
	Proper shipping name :	Not regulated as a dangerous good
	Packing group :	Not regulated as a dangerous good
Marine pollutant:		No
Transport regulations:		Transport this product in accordance with local/regional/national/international regulations.
Special precautions:		At the time of discharge from a tank truck, be sure to ground to prevent static electricity charge and to completely empty the remaining liquid in the pipes.

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Load the containers in such a way as not to fall down, tumble, or being damaged.

Cover the loaded cargo to prevent direct sunlight.

(Reference information) Heat accumulation storage test (UN recommendations) SAPT > 75°C (SAPT: Self-accelerating polymerization temperature)

15. Regulatory information

Inventories

Japan (ENCS) :	Listed
Korea (KECL) :	Listed
Australia (AIIC) :	Listed
Canada (DSL) :	Listed
China (IECSC) :	Listed
EU (EINECS) :	Listed
New Zealand (NZIoC) :	Listed
USA (TSCA) :	Listed
Philippines (PICCS) :	Listed
Taiwan (TCSI) :	Listed

Restriction

at export (Japan) : Not applicable

16. Other information

Disclaimer:

•The described information is based on the latest material, information, and other data available at the time of preparation; however, it does not intend to guarantee anything like contents, physical/chemical properties, and danger/hazards. Also, the precautions mentioned above are for ordinary handling; therefore, for special handling, it is the obligation of each user of the product to take proper measures suitable for each handling condition. Take enough care for such matters.

•This SDS is based on JIS Z7252: 2019 and JIS Z7253: 2019. If "Not classified" is provided in the GHS classification in Section 2, it means that sufficient information is available to classify the product and sufficient evidence is found to determine that it is not applicable to any hazard categories defined by JIS Z7252. And another case is that it is based on comprehensive judgment by experts.

•Regulatory information with regard to this substance in your country or region should be examined by your own responsibility.